Supporting Information

Directly converting syngas to linear α-olefins over core-shell $Fe_3O_4@MnO_2$ catalysts

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Table S1. Comparison of catalytic performances with the previous reports.

Catalyst	H_2/CO	T (°C)	P (MPa)	CO conv. (%)	CO_2 sel. (%)	CH_4 sel. (%)	Alkenes sel. (%)	LAOs/O[a] (%)	reactor	Ref.
Fe-Zn/2K,1Cu	2	270	2	60	43	6.1-6.9	NA[b]	63	Stirred reactor	[1]
Fe-based (with PEG Membrane)	0.67	270	3	20	34	3.9	NA	~69	fixed bed reactor	[2]
Fe-based (in PEG400 liquid)	1	240	3	26	NA	NA	NA	~71	slurry reactor	[3]
Pt (Co/Al_2O_3)	2	220	2	20	NA	~1	~33	39	milli-fixed bed reactor	[4]
$15Co3.7Mn/SiO_2$	2	260	1	19	3	4.5	~71	70	fixed bed reactor	[5]
$Co\text{-}SiO_2$	2	240	4.5	41	1.2	9.7	NA	41	fixed bed reactor	[6]
$Fe_2O_3@MnO_2$	1	280	2	89	42	10.3	69.1	NA	fixed bed reactor	[7]
$Fe_3O_4@MnO_2$	1	320	2	75	47	5.2	79.3	91	fixed bed reactor	[c]

[a] The ratio of linear α-$C_{4+}^{=}/C_{4+}^{=}$, LAOs stands for linear α-C_4^+ alkenes.

[b] Not available.

[c] This work.

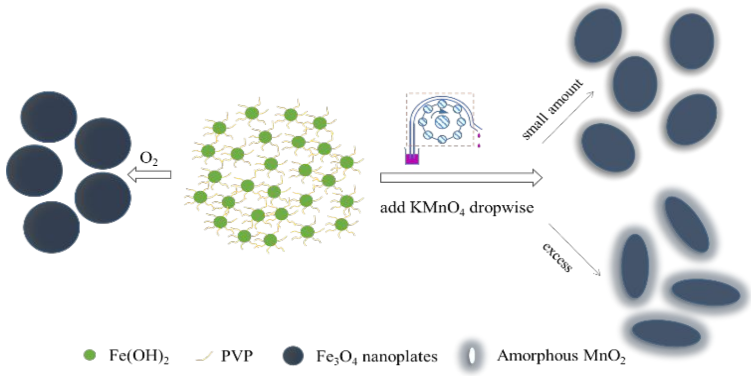
Table S2. The catalytic performance of Fe_3O_4@MnO_2 catalysts in the FTS reaction. [a]

Catalyst	Activity $[mol_{CO}g_{Fe}^{-1}s^{-1}]$	CO conv.(%)	CO_2 sel. (%)	Hydrocarbon sel. (%)			Alkenes sel. (%)		LAOs/O[b] (%)	α	C Balance (%)
				CH_4	C_2-C_3	C_{4+}	$C_{2+}^{=}$	$C_{4+}^{=}$			
Mn-control[c]	5.57E-05	87.58	46.94	18.96	30.58	50.46	42.75	25.29	50.12	0.61	95.86
Mn-free	2.22E-05	88.64	47.61	15.06	28.83	56.10	45.58	32.65	51.08	0.68	93.94
Mn-50	5.66E-05	67.90	47.11	3.58	14.63	81.79	79.34	65.81	87.10	0.83	92.35
Mn-60	6.11E-05	62.19	45.17	3.43	14.26	82.31	73.17	59.76	72.60	0.83	93.84

[a] Reaction conditions: catalyst (0.5g), syngas (CO:H_2=1:1), 2.0 MPa, 280°C. Pretreated with H_2 at 350°C for 10 h. The data is collected at 30 h.

[b] The ratio of linear α-$C_{4+}^{=}$/$C_{4+}^{=}$, LAOs stands for linear α-C_4^+ alkenes.

[c] The GHSV is changed to achieve the same FTY activity.



Scheme S1. Illustration of the preparation process of the Fe_3O_4 and $Fe_3O_4@MnO_2$.

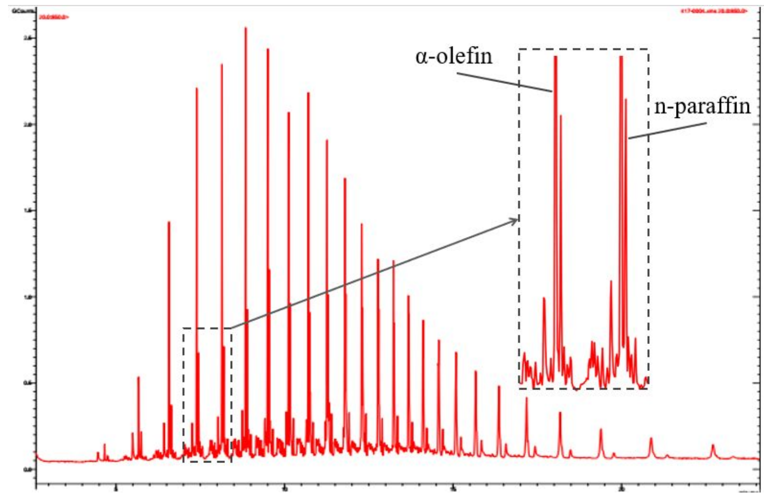


Figure S1. Typical GC-MS result of the liquid product.

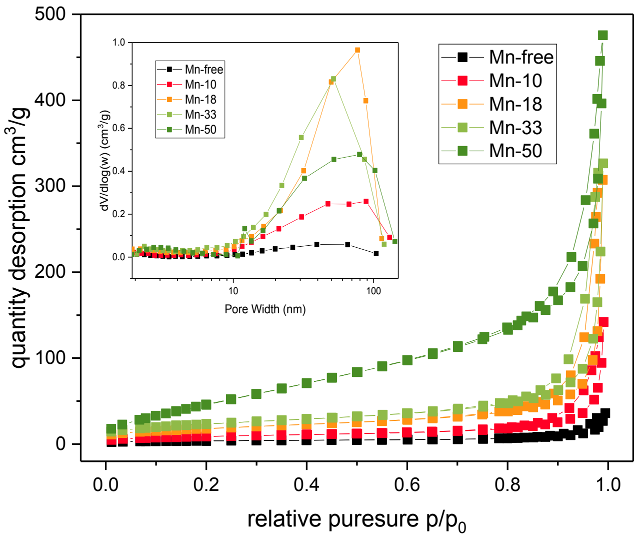


Figure S2. N_2 adsorption-desorption isotherms and BJH pore diameter distribution plots (inset) of Fe_3O_4@MnO_2 catalysts.

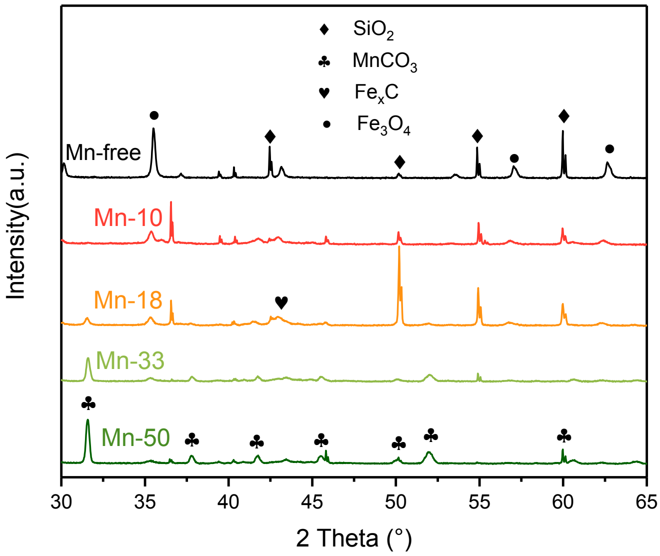


Figure S3. XRD patterns of the spent Fe_3O_4@MnO_2 catalysts (reaction conditions: 280 °C, 2.0 MPa, 3000 h^{-1}).

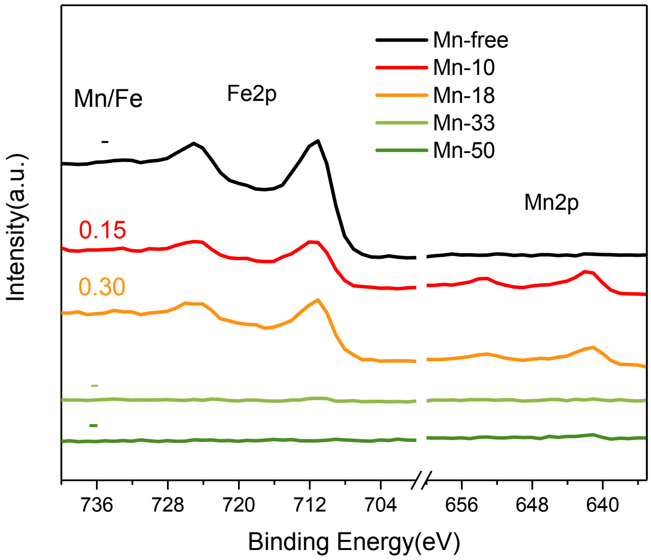


Figure S4. XPS spectra of the spent Fe_3O_4@MnO_2 catalysts (reaction conditions: 280 °C, 2.0 MPa, 3000 h^{-1}).

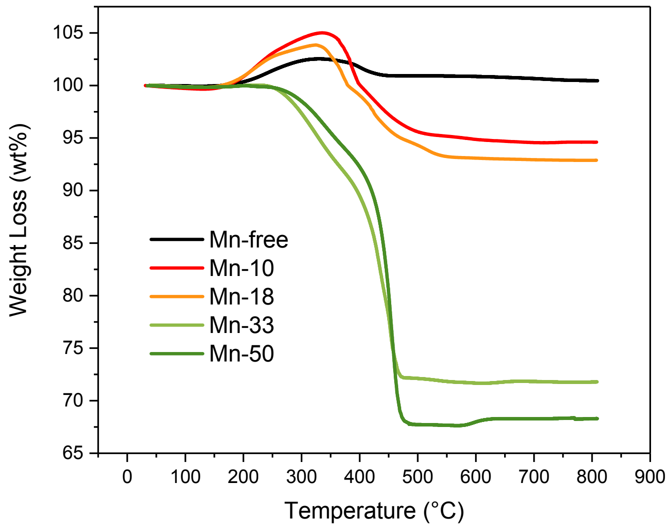


Figure S5. TGA results of the spent Fe_3O_4@MnO_2 catalysts in air condition.

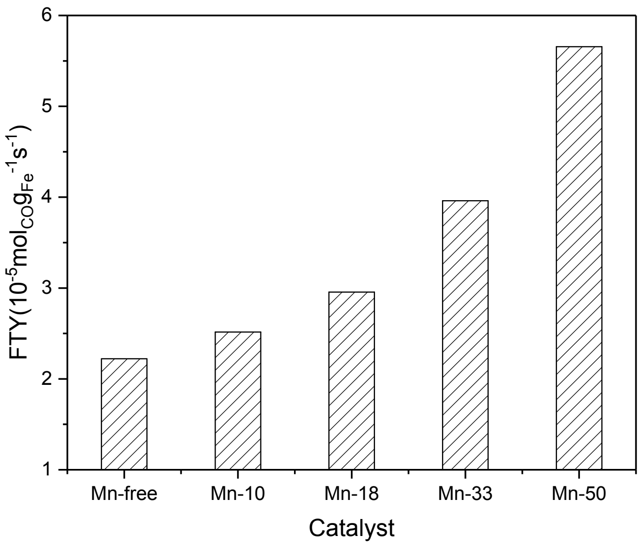


Figure S6. FTY activity of Fe_3O_4@MnO_2 catalysts. (Reaction conditions: 280°C, 2.0 MPa, 3000 h^{-1} and $H_2/CO = 1$).

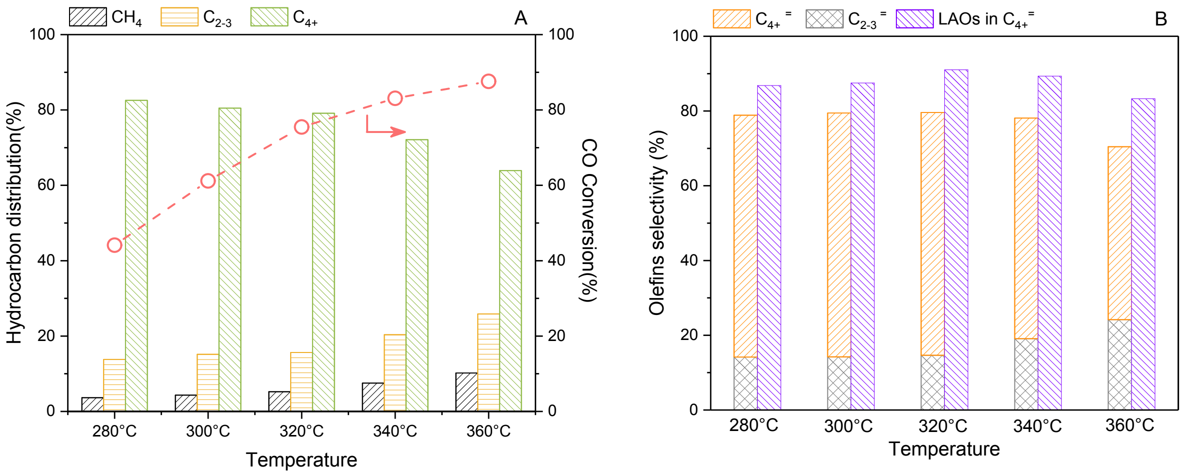


Figure S7. Effect of reaction temperature on (A) CO conversion and hydrocarbon distribution and (B) olefins and LAOs selectivity. Reaction conditions: P = 2.0 MPa, GHSV = 3000 h^{-1}, TOS = 10 h.

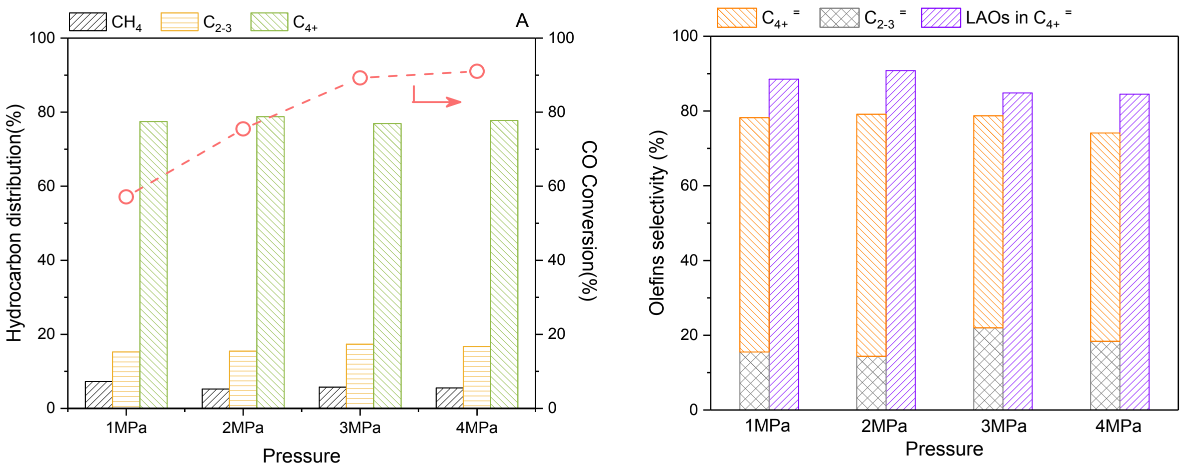


Figure S8. Effect of reaction pressure on (A) CO conversion and hydrocarbon distribution and (B) olefins and LAOs selectivity. Reaction conditions: T = 320 °C, GHSV = 3000 h^{-1}, TOS = 10 h.

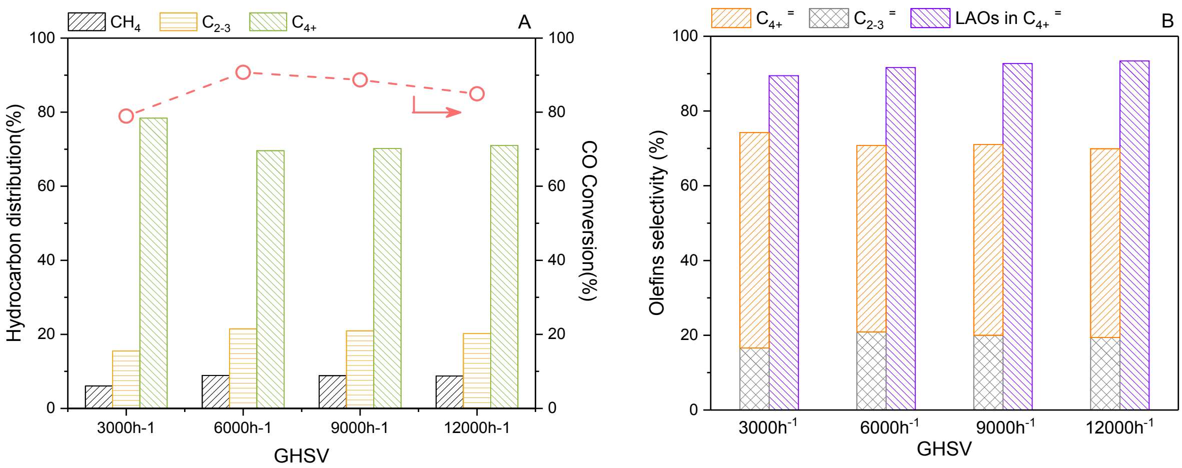


Figure S9. Effect of GHSV on (A) CO conversion and hydrocarbon distribution and (B) olefins and LAOs selectivity. Reaction conditions: T = 320 °C, P = 2.0 MPa, TOS = 10 h.

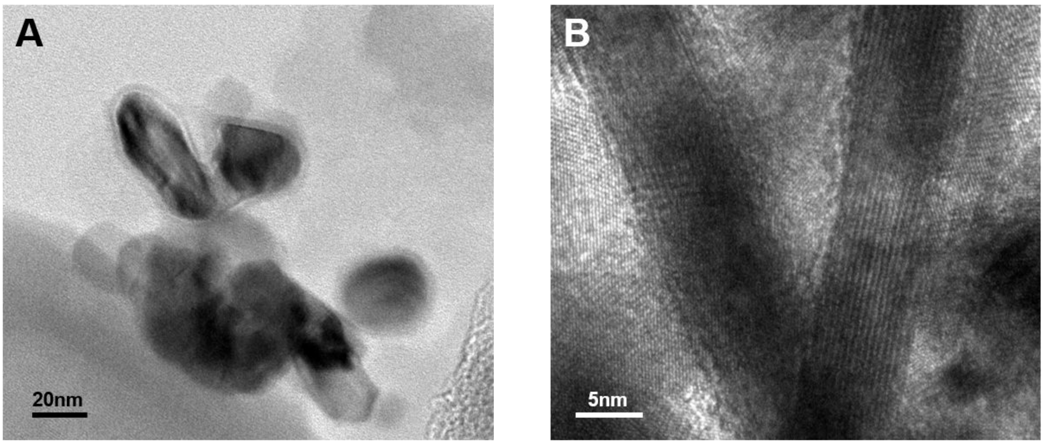


Figure S10. TEM images of spent Mn-50 catalyst after 30 h reaction.

Supplementary References

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